

HARSHAL JORWEKAR

SOFTWARE ENGINEER

+1 (669) 265-9125 | SAN JOSE, CA

jorwekar.h7@gmail.com | [linkedin.com/in/harshal-jorwekar](https://www.linkedin.com/in/harshal-jorwekar) | github.com/harshalJK

SUMMARY

Software Engineer with strong expertise in **full-stack** development, backend engineering, and **AI-driven** applications, skilled at building scalable data platforms, **ETL** pipelines, and **cloud-native** solutions. Experienced in **React**, **Next.js**, **Node.js**, and **Python**, with additional exposure to **Kotlin**, **Ruby**, and **Go**. Proficient in integrating **LLMs**, **embeddings**, and **generative AI** using frameworks like **LangChain**, **Hugging Face**, **TensorFlow**, and **PyTorch**, and deploying **AI agents** in production with workflow orchestration tools (**N8n**). Hands-on with multi-tenant **microservices**, event-driven architectures, and high-availability systems. Adept at collaborating with cross-functional teams to deliver secure, AI-enhanced, and **user-centric** solutions across finance, healthcare, and **SaaS** domains.

EDUCATION

Khoury College of Computer Sciences - Northeastern University

Master of Science in **Data Science**

Silicon Valley, CA

Sept 2023 - May 2025

Related Courses: Supervised and Unsupervised Machine Learning, Natural Language Processing (NLP), DBMS, Computer Vision

Savitribai Phule Pune University

Bachelor of **Computer Engineering**

Pune, IN

Aug 2017 - May 2021

Related Courses: Data Structures, Machine Learning, Mathematics, Algorithms, Artificial Intelligence, High Performance Computing

SKILLS

Programming Languages: Python, Java, Node.js, JavaScript, TypeScript, C++, C#, Go, Kotlin, Ruby, Bash, HTML, CSS, SQL

Cloud & Databases: AWS (DynamoDB, Athena, S3), GCP, Azure, PostgreSQL, MongoDB, SQL Server, MySQL, Oracle, Snowflake, Databricks, Redshift, Starburst, Vector Databases (FAISS, Pinecone, Weaviate)

Frameworks & Tools: React, Next.js, React Native, Angular, Flask, FastAPI, .NET Core, Micronaut, jOOQ, Docker, Kubernetes, Jenkins, Git, Terraform, Kafka, Spark, N8n, LangChain

AI/ML & Data: PyTorch, TensorFlow, Hugging Face, OpenAI APIs, Generative AI, Embeddings, RAG systems, ETL pipelines, Knowledge Graphs, Data Governance & Access Controls

Testing & Practices: Cypress, Jest, Playwright, Unit Testing, Agile, Scrum, CI/CD, REST APIs, Microservices, Event-driven Architectures, OOD

WORK EXPERIENCE

Fidelity Information Services (Pune, IN)

Aug 2021 - July 2023

Software Engineer II (Offshore Team - USA Client)

- Designed and optimized scalable backend **APIs** and **ETL** pipelines for fintech applications, enabling **30%** faster transaction processing and secure **data** access across multi-tenant **microservices**.
- Implemented data governance and **access control mechanisms** ensuring compliance with privacy standards and reducing audit risks.
- Developed event-driven services leveraging **Kafka** to handle high-volume, real-time financial **transactions**.
- Enhanced **CI/CD** pipelines with **Jenkins**, **Docker**, and **Kubernetes**, achieving **40%** fewer defects and faster release cycles.
- Collaborated with product teams to deploy **AI-assisted fraud detection models** in production, integrating TensorFlow and PyTorch for anomaly detection.

Avirna Ventures (Pune, IN)

Aug 2020 - July 2021

Software Engineering Intern

- Built responsive **full-stack** web apps with **React**, **Node.js**, and **PostgreSQL**, improving engagement by **25%**.
- Integrated workflow automation using **Node.js** orchestration tools to streamline data ingestion and reporting.
- Assisted in implementing unit and **end-to-end testing** with Jest and Playwright, improving test coverage.
- Ensured compliance with healthcare data governance and **HIPAA security** policies through secure coding practices.

PROJECTS

SANSKRITA – Retrieval-Augmented LLM for Sanskrit Q&A (Capstone)

Jan 2024 - May 2025

- Designed a modular **RAG** pipeline with **LangChain**, **Hugging Face**, and **PyTorch**, fine-tuned **GPT-2** on IAST-transliterated Sanskrit corpora.
- Integrated **FAISS** and vector databases (**Pinecone**, **Weaviate**) for **semantic** retrieval, reducing hallucinations and improving factual grounding.
- Deployed system on **AWS/GCP** with **Next.js** + **React frontend**, supporting AI-driven search across large corpora.

Multi-Modal Dynamic Hand Gesture Recognition System

Sept 2024 - Dec 2024

- Developed a real-time AI system integrating **RGB**, **depth**, and **EMG** data streams using an event-driven, **multi-stream** neural architecture.
- Achieved **97.25%** accuracy and **<40ms latency**, suitable for **AR/VR** and prosthetics.
- Packaged models as **microservices** with **Kubernetes**, enabling scalable deployment and **cloud-based** inference pipelines.